

Si Liu

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Research Interests

Si Liu's research lies at the intersection of **formal methods**, **databases**, and **distributed systems**, with a strong focus on **leveraging formal verification techniques to build reliable, secure, and performant systems**. In recent years, he has worked on verifying and validating both the *designs and deployments* of databases systems and DNS, addressing not only *qualitative* (e.g., reliability and security) but also *quantitative* (e.g., availability and scalability) aspects. He is also interested in the intersection of these areas with AI: both in applying AI techniques to strengthen verification, and in developing rigorous methods for verifying AI-powered systems themselves.

Education

University of Illinois Urbana-Champaign (UIUC)

PHD IN COMPUTER SCIENCE

Aug. 2012 – May 2019

East China Normal University (ECNU)

MASTER IN COMPUTER SCIENCE

Sep. 2009 – May 2012

East China Normal University (ECNU)

BACHELOR IN SOFTWARE ENGINEERING

Sep. 2005 – Jun. 2009

Professional Experience

Aug. 2025 – present	Tenure-track Assistant Professor , Texas A&M University
May 2023 – July 2025	Senior Researcher (Deutsch: Oberassistent) , ETH Zurich
Aug. 2019 – Apr. 2023	Postdoc Researcher , ETH Zurich
Oct. 2014 – Sep. 2018	Research Assistant , Assured Cloud Computing Center, UIUC
Jun. 2011 – Aug. 2011	Research Intern , National Cyber-Physical Systems Camp, USTC
Sep. 2009 – May 2012	Research Assistant , Danish-Chinese Research Center, ECNU
May 2008 – Feb. 2009	Software Engineering Intern , Dept. of Talent House, Hewlett-Packard Co. Ltd.

Awards & Honors

Mar. 2024 – Mar. 2027	Official Project Partner on the grant for Project “Formal Verification of Isolation Guarantees in Database Systems” , SNSF (Swiss National Science Foundation)	CHF 181,099
Aug. 2024 – July 2025	Career Seed Award , ETH Zurich	CHF 30,000
2024	Thesis supervisor for Luca Multazz, recipient of the ETH Medal for his Master thesis (Top 2.5%) , ETH Zurich	
2023	Thesis supervisor for Lukas Heimes, recipient of the ETH Medal for his Master thesis (Top 2.5%) , ETH Zurich	
2012	Outstanding Graduate Award (Ranked 1st) , East China Normal University	
2011	Excellent Student Award (Ranked 1st) , East China Normal University	

Publications

* I contributed equally as a co-first author † indicates the students I supervised

● FM/SE x DATABASES (RECENT)

- [FM’26] **Si Liu***, Ziwei Zhou*, Zhou Zhou, Peixin Wang, Min Zhang. A Formal Framework for Predicting Distributed System Performance under Faults
- [VLDB’26] Zhiheng Cai†, **Si Liu**, Hengfeng Wei, Yuxing Chen, Anqun Pan. Fast Verification of Strong Database Isolation
- [SIGMOD’26] Zijong Yin†, **Si Liu**, David Basin. Testing Graph Databases with Synthesized Queries
- [VLDB’25] Shabnam Ghasemirad†, **Si Liu**, Christoph Sprenger, Luca Multazzu, David Basin. VerIso: Verifiable Isolation Guarantees for Database Transactions
- [ICDE’25] Hengfeng Wei, Jiang Xiao, Na Yang, **Si Liu**, Zijong Yin, Yuxing Chen, Anqun Pan. Boosting End-to-End Database Isolation Checking via Mini-Transactions
- [TACAS’25] Shabnam Ghasemirad†, Christoph Sprenger, **Si Liu**, Luca Multazzu, David Basin. Pushing the Limit: Verified Performance-Optimal Causally Consistent Database Transactions
- [OOPSLA’24] **Si Liu**, Long Gu, Hengfeng Wei, David Basin. Plume: Efficient and Complete Black-box Checking of Weak Isolation Levels
- [SIGMOD’24] **Si Liu**, Luca Multazzu, Hengfeng Wei, David Basin. NOC-NOC: Towards Performance-optimal Distributed Transactions
- [VLDB’24] Long Gu†, **Si Liu**, Tiancheng Xing, Hengfeng Wei, Yuxing Chen, David Basin. IsoVista: Black-box Checking Database Isolation Guarantees. **Demo Track**
- [VLDB’23] **Si Liu***, Kaile Huang*, Zhenge Chen, Hengfeng Wei, David Basin, Haixiang Li, Anqun Pan. Efficient Black-box Checking of Snapshot Isolation in Databases
- [OSDI’23] Zu-Ming Jiang, **Si Liu**, Manuel Rigger, Zhendong Su. Detecting Transactional Bugs in Database Engines via Graph-Based Oracle Construction
- [OOPSLA’22] **Si Liu**, Jose Meseguer, Peter Csaba Olveczky, Min Zhang, David Basin. Bridging the Semantic Gap between Qualitative and Quantitative Models of Distributed Systems
- [TOSEM’22] **Si Liu**. All in One: Design, Verification, and Implementation of SNOW-Optimal Read Atomic Transactions

• FM x SECURITY (DNS AND BEYOND)

- [POPL’25] Dhruv Nevatia†, **Si Liu**, David Basin. Reachability Analysis of the Domain Name System.
- [USENIX SEC’24] Huayi Duan, Marco Bearzi, Jodok Vieli, Adrian Perrig, David Basin, **Si Liu**, Bernhard Tellenbach. CAMP: Compositional Amplification Attacks against DNS
- [SIGCOMM’23] **Si Liu**, Huayi Duan, Lukas Heimes, Marco Bearzi, Jodok Vieli, Adrian Perrig, David Basin. A Formal Framework for End-to-End DNS Resolution
- [NSDI’23] Huayi Duan, Fischer Ruben, Lou Jie, **Si Liu**, David Basin, Adrian Perrig. RHINE: Robust and High-performance Internet Naming with E2E Authenticity
- [CSF’22] **Si Liu***, Thilo Weghorn*, Christoph Sprenger, Adrian Perrig, David Basin. N-Tube: Formally Verified Secure Bandwidth Reservation in Path-Aware Internet Architectures
- [Book Chapter] David Basin, Tobias Klenze, **Si Liu**, Christoph Sprenger. *Design-Level Verification in The Complete Guide to SCION: From Design Principles to Formal Verification*. 2022
- [Book Chapter] Giacomo Giuliani, Markus Legner, **Si Liu**, Adrian Perrig, Thilo Weghorn, Marc Wyss. *Extensions for the Data Plane in The Complete Guide to SCION: From Design Principles to Formal Verification*. 2022

• OTHER PUBLICATIONS

- [ICASSP’26] Shi Peng, **Si Liu**, Peixin Wang, Dapeng Zhi, Min Zhang. Identifying Critical States via Action Consensus.
- [NN’26] Shi Peng, **Si Liu**, Dapeng Zhi, Peixin Wang, Min Zhang. MaskCtrl: Training Mask Networks as Self-Explainable and Performant Controllers via Deep Reinforcement Learning. *Neural Networks*, 2026

- [NN'25] Shi Peng, **Si Liu**, Dapeng Zhi, Peixin Wang, Chenyang Xu, Cheng Chen, Min Zhang. ATA: An Abstract-Train-Abstract approach for explanation-friendly deep reinforcement learning. *Neural Networks*, 2025
- [CAV'24] Dapeng Zhi, Peixin Wang, **Si Liu**, Luke Ong, Min Zhang. Unifying Qualitative and Quantitative Safety Verification of DNN-Controlled Systems
- [VMCAI'24] Jiaxu Tian, Dapeng Zhi, **Si Liu**, Peixin Wang, Guy Katz, Min Zhang. Taming Reachability Analysis of DNN-Controlled Systems via Abstraction-Based Training
- [NeurIPS'23] Jiaxu Tian, Dapeng Zhi, **Si Liu**, Peixin Wang, Cheng Chen, Min Zhang. Boosting Verification of Deep Reinforcement Learning via Piece-Wise Linear Decision Neural Networks
- [CVPR'23] Zhaodi Zhang, Zhiyi Xue, Yang Chen, **Si Liu**, Yueling Zhang, Jing Liu, Min Zhang. Boosting Verified Training for Robust Image Classifications via Abstraction
- [ISSTA'23] Zhiyi Xue, **Si Liu**, Zhaodi Zhang, Yiting Wu, Min Zhang. A Tale of Two Approximations: Tightening Over-Approximation for DNN Robustness Verification via Under-Approximation
- [ASE'22] Zhaodi Zhang, Yiting Wu, **Si Liu**, Jing Liu, Min Zhang. Provably Tightest Linear Approximation for Robustness Verification of Sigmoid-like Neural Networks
- [TASE'21] Lei Liang, **Si Liu**. Exploring Design Alternatives for Replicated RAMP Transactions Using Maude
- [NFM'20] **Si Liu**, Atul Sandur, Jose Meseguer, Peter Olveczky, Qi Wang. Generating Correct-by-Construction Distributed Implementations from Formal Maude Designs
- [TACAS'19] **Si Liu**, Peter Csaba Olveczky, Min Zhang, Qi Wang, Jose Meseguer. Automatic Analysis of Consistency Properties of Distributed Transaction Systems in Maude.
- [FAoC'19] **Si Liu**, Peter Csaba Olveczky, Qi Wang, Indranil Gupta, José Meseguer. Read Atomic Transactions with Prevention of Lost Updates: ROLA and Its Formal Analysis. *Formal Aspects of Computing*
- [CCS'19] Qi Wang, Pubali Datta, Wei Yang, **Si Liu**, Carl Gunter, Adam Bates. Charting the Attack Surface of Trigger-Action IoT Platforms.
- [FASE'18] **Si Liu**, Peter Csaba Olveczky, Keshav Santhanam, Qi Wang, Indranil Gupta, José Meseguer. ROLA: A New Distributed Transaction Protocol and Its Formal Analysis.
- [LITES'17] **Si Liu**, Jatin Ganhotra, Muntasir Raihan Rahman, Son Nguyen, Indranil Gupta, José Meseguer. Quantitative Analysis of Consistency in NoSQL Key-value Stores. *Leibniz Transactions on Embedded Systems*
- [ICFEM'17] **Si Liu**, Peter Csaba Olveczky, Jatin Ganhotra, Indranil Gupta, José Meseguer. Exploring Design Alternatives for RAMP Transactions through Statistical Model Checking.
- [JLAMP'16] **Si Liu**, Peter Csaba Olveczky, José Meseguer. Modeling and Analyzing Mobile Ad hoc Networks in Real-Time Maude. *Journal of Logical and Algebraic Methods in Programming*
- [SAC'16] **Si Liu**, Peter Csaba Olveczky, Muntasir Raihan Rahman, Jatin Ganhotra, Indranil Gupta, José Meseguer. Formal Modeling and Analysis of Ramp Transaction Systems.
- [QEST'15] **Si Liu**, Son Nguyen, Jatin Ganhotra, Muntasir Raihan Rahman, Indranil Gupta, José Meseguer. Quantitative Analysis of Consistency in NoSQL Key-value Stores. *Nominated for Best Paper*.
- [ICFEM'14] **Si Liu**, Muntasir Raihan Rahman, Stephen Skeirik, Indranil Gupta, José Meseguer. Formal Modeling and Analysis of Cassandra in Maude.
- [PRDC'14] Xi Wu, **Si Liu**, Huibiao Zhu and Yongxin Zhao. Reasoning about Group-Based Mobility in MANETs.
- [ComSIS'13] Xi Wu, Huibiao Zhu, Yongxin Zhao, Zheng Wang, **Si Liu**. Modeling and verifying the Ariadne protocol using process algebra. *Computer Science and Information Systems Journal*
- [ECBS'12] Xi Wu, **Si Liu**, Huibiao Zhu, Yongxin Zhao, Lei Chen. Modeling and Verifying the Ariadne Protocol Using CSP.
- [HASE'11] **Si Liu**, Yongxin Zhao, Huibiao Zhu, Qin Li. A Calculus for Mobile Ad Hoc Networks from a Group Probabilistic Perspective.
- [SSIRI'11] **Si Liu**, Xiaofeng Wu, Qin Li, Huibiao Zhu, Qian Wang. Formal Approaches to Wireless Sensor Networks.
- [TASE'11] **Si Liu**, Yongxin Zhao, Huibiao Zhu, Qin Li. Towards a Probabilistic Calculus for Mobile Ad Hoc Networks.

- [TASE'11]** Mengying Wang, Huibiao Zhu, Yongxin Zhao, **Si Liu**. Modeling and Analyzing the μ TESLA Protocol Using CSP.
- [ICECCS'11]** Yongxin Zhao, Yanhong Huang, Jifeng He, **Si Liu**. Formal Model of Interrupt Program from a Probabilistic Perspective.
- [UTP'10]** Qin Li, Yongxin Zhao, Xiaofeng Wu, **Si Liu**. Promoting Models.

WORKSHOP PAPERS

- [WRLA'18]** **Si Liu**, Peter Csaba Ölveczky, Qi Wang, José Meseguer. Formal Modeling and Analysis of the Walter Transactional Data Store.
- [SSS'15]** **Si Liu**, Peter Csaba Ölveczky and José Meseguer. Formal analysis of Leader Election in MANETs Using Real-Time Maude.
- [WRLA'14]** **Si Liu**, Peter Csaba Ölveczky and José Meseguer. A Framework for Mobile Ad hoc Networks in Real-Time Maude.
- [SSIRI'11]** Han Zhu, Huibiao Zhu, **Si Liu**, Jian Guo. Towards Denotational Semantics for Verilog in PVS.

OTHER BOOK CHAPTERS

- Rakesh Bobba, Jon Grov, Indranil Gupta, **Si Liu**, José Meseguer, Peter Csaba Ölveczky, Stephen Skeirik. Survivability: Design, Formal Modeling, and Validation of Cloud Storage Systems using Maude. 2018.

PHD THESIS

- Design, Verification and Automatic Implementation of Correct-by-Construction Distributed Transaction Systems in Maude. University of Illinois Urbana-Champaign. 2019

Teaching Experience

Fall 2024	Computer Systems , Teaching Assistant	ETH Zurich
Spring 2024	Data Modeling and Databases , Teaching Assistant	ETH Zurich
Fall 2023	Computer Systems , Teaching Assistant	ETH Zurich
Spring 2023	Data Modeling and Databases , Teaching Assistant	ETH Zurich
Fall 2022	Information Systems for Engineers , Head Teaching Assistant	ETH Zurich
Spring 2022	Data Modeling and Databases , Teaching Assistant	ETH Zurich
Fall 2021	Applied Security Lab , Teaching Assistant	ETH Zurich
Spring 2021	Data Modeling and Databases , Teaching Assistant	ETH Zurich
Fall 2020	Computer Systems , Teaching Assistant	ETH Zurich
Spring 2020	Data Modeling and Databases , Teaching Assistant	ETH Zurich
Fall 2019	Applied Security Lab , Teaching Assistant	ETH Zurich
Fall 2016	Distributed Systems , Teaching Assistant	UIUC
Fall 2011	Process Algebra , Teaching Assistant	ECNU
Spring 2011	Algorithms and Data Structures , Teaching Assistant	ECNU
Fall 2010	Discrete Mathematics , Teaching Assistant	ECNU
Spring 2010	Algorithms and Data Structures , Teaching Assistant	ECNU

Mentoring

- I am fortunate to have the opportunity to work with the following talented students:
- **PhD students (10 in total)**
 - Dhruv Nevatia (co-advising with Prof. David Basin), ETH Zurich
 - Shabnam Ghasemirad (co-advising with Dr. Christoph Sprenger and Prof. David Basin), ETH Zurich

- Zijong Yin, ETH Zurich
- Ziwei Zhou, ECNU
- Shi Peng, ECNU
- Zhaodi Zhang, ECNU
- Zhou Zhou, ECNU,
- Zhiheng Cai, Tsinghua University
- Qiuhan Xiong, Nanjing University ECNU
- Qikang Liu, Simon Fraser University

• **Graduate students (21 in total, including two who were awarded the ETH Medal)**

- Kumaran Gowrisankar, Texas A&M University
- Sahithi Duppati, Texas A&M University
- Jaelen Dixon, Texas A&M University
- Shashank Chandavarkar, Texas A&M University
- Wenyuan Jiang, ETH Zurich
- Rongchuan Liu, ETH Zurich
- Theodor Moroianu, ETH Zurich
- Yunxin Sun, ETH Zurich
- Yufei Zhang, ETH Zurich
- Rolando Grave de Peralta, ETH Zurich
- Zhou Zhou, ECNU
- Long Gu, Nanjing University
- Qiuhan Xiong, Nanjing University
- Luca Multazz, ETH Zurich

* **honoured with the ETH Medal for his Master thesis “NOCS-Optimal Distributed Transactions and Beyon” (top 2.5%)**

- Lukas Heimes, ETH Zurich

* **honoured with the ETH Medal for his Master thesis “A Formal Framework for End-to-End DNS Resolution” (top 2.5%)**

- Jodok Vieli, ETH Zurich
- Marco Bearzi, ETH Zurich
- Shabnam Ghasemirad (now PhD at ETH Zurich), ETH Zurich
- Jiang Xiao, Nanjing University
- Jiaxu Tian, ECNU
- Zhiyi Xue, ECNU

- **Undergraduate students (11 in total)**

- Bosheng Peng, Nanjing University
- Tiancheng Xing, Nanjing University
- Zhiheng Cai, Nanjing University
- Long Gu, Nanjing University
- Zhenge Chen, Nanjing University
- Zhou Zhou, ECNU
- Ziwei Zhou, ECNU
- Ruiyang Liu, ECNU
- Lei Liang, ECNU
- Plamen Stefanov, ETH Zurich
- Keshav Santhanam, UIUC

Professional Activities

- **Organizer:** SCCP'26 (Symposium on Checking Consistency Principles)
- **Program Committee:** SCCP'26, DASFAA'26, NSDI'25, ICFEM'24, ICFEM'23, ICFEM'22
- **Journal Reviewer:** The VLDB Journal, Journal of Logical and Algebraic Methods in Programming, Autonomous Agents and Multi-Agent Systems, Theory of Computing Systems
- **Sub-Reviewer:** ICFEM ('16, '14, '13, '12, '11, '10), FM ('21, '18), FASE'14, TASE ('12, '11)
- **Judge:** Student Research Competition (SPLASH 2024), Texas Junior Academy of Science (TJAS) 2025
- **Dagstuhl Seminar:** “Ensuring the Reliability and Robustness of Database Management Systems (21442)” 2021 (participant)
- **Grant:**
 - Official Project Partner on the SNSF grant for Project “Formal Verification of Isolation Guarantees in Database Systems” (CHF 181,099; 2024–2027)
 - Preparing the proposal for NSF CNS 1409416 (\$584,508; 2014–2018), Availability-Consistency Trade-offs in Key-Value and NoSQL Storage Systems

Invited Talks & Presentations

- Making Database Transactions Reliable and Performant: from Design to Deployment. Department of Computer Science, Aarhus University, 2025.
- Making Database Transactions Reliable and Performant: from Design to Deployment. Department of Computer Science, UC Irvine, 2025.
- Making Database Transactions Reliable and Performant: from Design to Deployment. Department of Computer Science and Engineering, Texas A&M University, 2025.
- Making Database Transactions Reliable and Performant: from Design to Deployment. Department of Computer Science, UT Dallas, 2025.

- Making Database Transactions Reliable and Performant: from Design to Deployment. School of Computer Science and Engineering, UNSW Sydney, 2025.
- Making Database Transactions Reliable and Performant: from Design to Deployment. School of Computer Science, University of Sydney, 2025.
- Making Database Transactions Reliable and Performant: from Design to Deployment. School of Data Science, The Chinese University of Hong Kong (Shenzhen), 2025.
- Making Distributed Systems Dependable and Performant: from Design to Deployment. School of Computer Science and Technology, Nanjing University, 2024.
- Plume: Efficient and Complete Black-box Checking of Weak Isolation Levels. OOPSLA'24, Pasadena, USA, 2024.
- NOC-NOC: Towards Performance-optimal Distributed Transactions. SIGMOD'24, Santiago, Chile, 2024.
- Generating Correct-by-Construction Distributed Implementations from Formal Maude Designs. NFM'20, Virtual, 2020.
- Design, Verification and Automatic Implementation of Correct-by-Construction Distributed Transaction Systems in Maude. University Paris Diderot, France, 2019.
- Design, Verification and Automatic Implementation of Correct-by-Construction Distributed Transaction Systems in Maude. ETH Zurich, Switzerland, 2019.
- Automatic Analysis of Consistency Properties of Distributed Transaction Systems in Maude. TACAS'19, Prague, Czech Republic, 2019.
- Exploring Design Alternatives for RAMP Transactions through Statistical Model Checking. ICFEM'17, Xi'an, China, 2017.
- Design, Formal Modeling, and Validation of Cloud Storage Systems using Maude. Huawei, Urbana-Champaign, USA, 2017.
- Exploring Design Alternatives for the RAMP Transaction System Through Statistical Model Checking. Assured Cloud Computing Center, Urbana-Champaign, USA, 2017.
- Formal Modeling and Analysis of Ramp Transaction Systems. SAC'16, Pisa, Italy, 2016.
- Formal Modeling and Analysis of Ramp Transaction Systems. Assured Cloud Computing Center, Urbana-Champaign, USA, 2016.
- Quantitative Analysis of Consistency in NoSQL Key-value Stores. QEST'15, Madrid, Spain, 2015.
- Quantitative Analysis of Consistency in NoSQL Key-value Stores. Assured Cloud Computing Center, Urbana-Champaign, USA, 2015.
- A Framework for Mobile Ad hoc Networks in Real-Time Maude. WRLA'14, Grenoble, France, 2014.
- Formal Modeling and Analysis of Cassandra in Maude. Assured Cloud Computing Center, Urbana-Champaign, USA, 2014.
- A Calculus for Mobile Ad Hoc Networks from a Group Probabilistic Perspective. HASE'11, Boca Raton, USA, 2011.
- Towards a Probabilistic Calculus for Mobile Ad Hoc Networks. USTC, Suzhou, China, 2011.
- Towards a Probabilistic Calculus for Mobile Ad Hoc Networks. TASE'11, Xi'an, China, 2011.
- Formal Approaches to Wireless Sensor Networks. SSIRI'11, Jeju Island, Korea, 2011.

Referees

Prof. José Meseguer

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¹Ms. Allison Mette is managing emails on behalf of Prof. José Meseguer, and this is her email address (along with her phone number).